



CSC104-Database Systems
Mid Project Report(Spring 2025)



Student Details

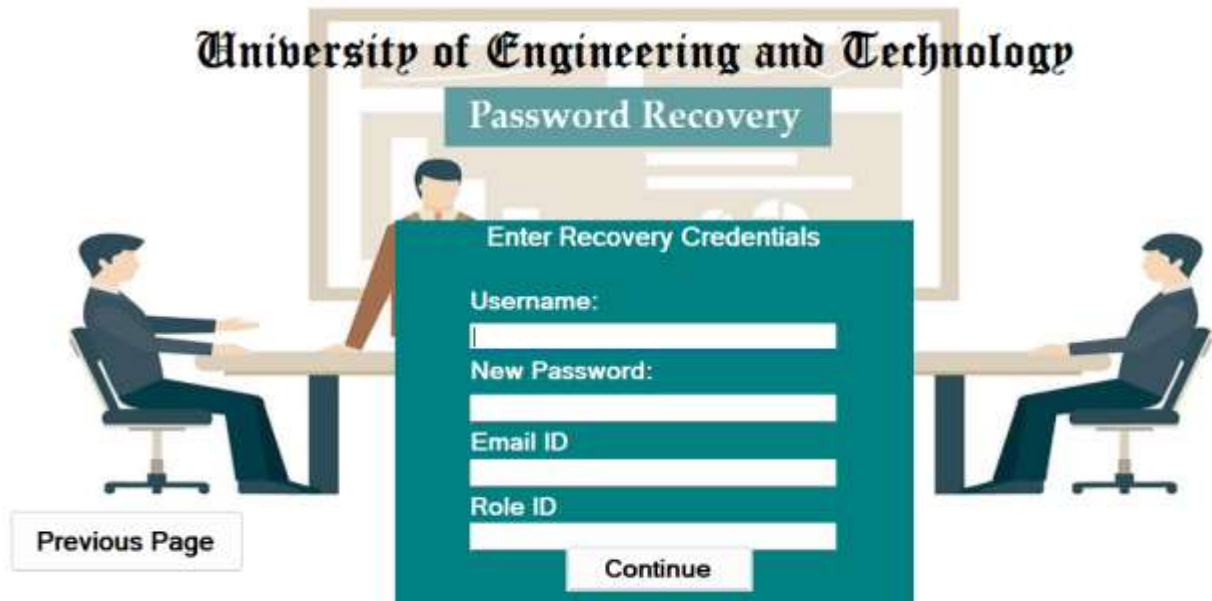
Repository Name	DBS25P211
Registration Number	2024-CS-211

Project	
Project Title	Faculty Management System
Executive Summary	<i>The Faculty Management System is designed to help the university efficiently manage faculty workload and resource allocation. These tasks—such as assigning courses, tracking faculty responsibilities, and handling resource requests (e.g., board markers, consumables)—were managed manually using spreadsheets and emails. This manual process often led to errors, delays, and inefficiencies, making it difficult to track workload distribution and resource availability. With this system, everything is organized in one place. Faculty members log in securely, and their access is based on their role. Faculty can view assigned courses, submit requests for resources, and check their request status. The Department Head oversees the system, ensuring that courses are assigned fairly, faculty requests are reviewed, and resources are distributed efficiently.</i> <i>When a faculty member needs materials such as board markers, stationery, they can submit a request through the system. The request is marked as Pending until reviewed by the Department Head. Once approved, the requested resource is allocated, and the status is updated to Fulfilled. Faculty members can easily track their requests without needing multiple follow-ups via email.</i> <i>The system also ensures that faculty workload is evenly distributed. It monitors teaching assignments, research supervision, and administrative responsibilities, preventing any faculty member from being overburdened while others have fewer tasks. Reports can be generated using the given system which will help the department plan better, ensuring smooth operations and compliance with academic requirements. Overall, the system enhances transparency, efficiency, and better resource management for the university.</i>

Interface Name		Roll Selector	
Picture			
			
UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Role Selector	Combo Box	Button1_click	Lookup
Continue	Button	Button1_click	

Interface Name		Login Page	
Picture			
			
UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table

Username	Textbox	CheckCred	Users.username
Password	Textbox	Hash password	Users.passwrod_hash
Forgot Password	Linked Label		
Continue	Button		

Interface Name	Password Recovery		
Picture			
UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Username	Textbox	Update Password	Users.username
Password	Textbox	Hash Password	Users.password_hash
Email ID	Textbox	Update Password	Users.email
Role ID	Textbox	Update Password	Users.role_id
Continue	Button		

Interface Name		Add Profiles (by admin staff)
Picture		

Univerisity of Engineering and Technology Lahore

Admin Staff

Faculty Management

View All Profiles

Add Profile

Edit Profile

Delete Profile

Previous Page

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
View All Profiles	Button	ShowUsers	Faculty
Add profiles	Button	FacCrud	Users, Faculty
Edit Profiles	Button	FacCrud	Users, Faculty
Deletes Profile	Button	FacCrud	Users, Faculty
Previous Page	Button		

Interface Name

Faculty Requests Processing

Picture

Univerisity of Engineering and Technology Lahore

Admin Staff

Requeset Processing

faculty_id	name	request_id	item_id	quantity
1				

Previous Page

Enter Faculty Request ID

Fulfill

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Request Processing	DataGridView	ProcessReq	Faculty_requests

Interface Name	Update User
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Picture

University of Engineering and Technology Lahore

Admin Staff

Faculty Management

Enter Credentials

Existing User ID:

Username:

Password:

Email ID:

Existing Faculty ID:

Contact:

Research Area:

Total Teaching Hours:

Continue

Previous Page

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
User ID	TextBox	FacCrud	Users.user_id
Username	TextBox	FacCrud	Users.username
Password	TextBox	FacCrud	Users.password_hash
Email	TextBox	FacCrud	Users.email
Faculty Id	TextBox	FacCrud	Faculty.faculty_id
Contact	TextBox	FacCrud	Faculty.Contact
Research Area	TextBox	FacCrud	Faculty.reseacrh_area
Total Teaching Hours	TextBox	FacCrud	Faculty.Total_teaching_hours
Continue	Button		

Interface Name	Room Management
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Picture

University of Engineering and Technology Lahore

Admin Staff

Room Management

View All Rooms

Add Room

Edit Rooms

Delete a Room

Previous Page

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Add Room	Button	RoomCrud	Rooms
Delete Room	Button	RoomCrud	Rooms
Update Room	Button	RoomCrud	Rooms
View All Rooms	Button	ShowRooms	Rooms
Previous Page	Button		

Interface Name

Course Management

Picture

University of Engineering and Technology Lahore

Admin Staff

Course Management

View All Courses

Add Course

Update Course

Delete Course

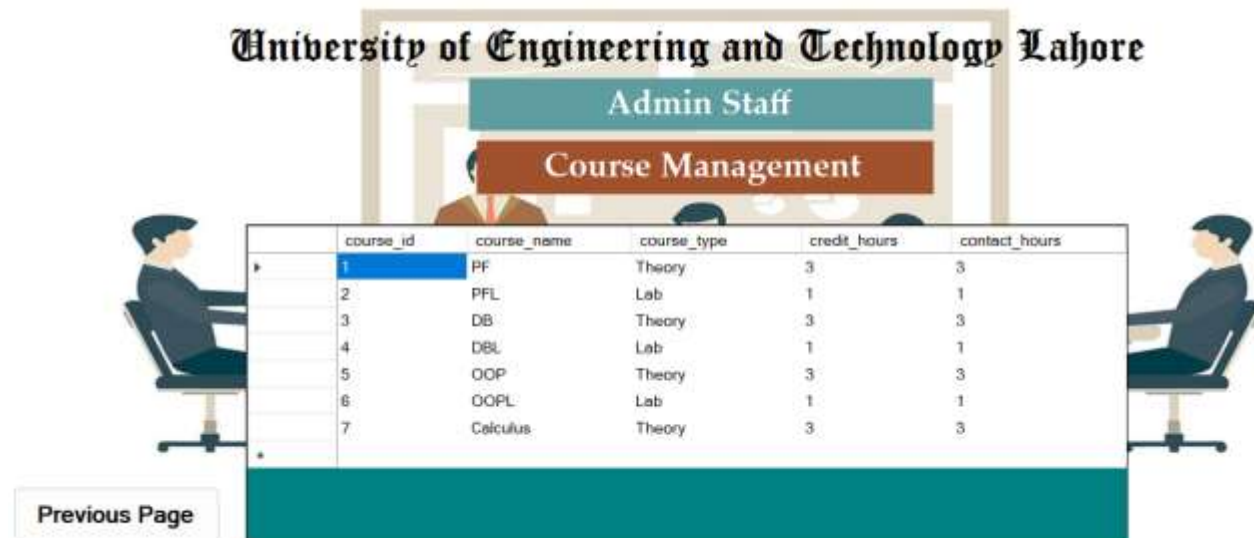
Previous Page

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
View All Courses	Button	ShowCourses	Courses
Add Course	Button	RoomCrud	Courses
Update Course	Button	RoomCrud	Courses
Delete Course	Button	RoomCrud	Courses
Previous Page	Button		

Interface Name

View All Courses

Picture



UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Course View	DataGridView	ShowCourses	Courses

Interface Name

Project Management

Picture

University of Engineering and Technology Lahore

Admin Staff

Project Management

View All Projects

Add Project

Edit Project

Delete a Project

Previous Page

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
View All Projects	Button	ShowProjects	Projects
Add Projects	Button	ProjCrud	Projects
Edit Project	Button	ProjCrud	Projects
Delete Project	Button	ProjCrud	Projects
Previous Page	Button		

Interface Name

Course Assignment

Picture

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Department Head

Workload Management

Assign Course

Edit

Delete Course

View Assigned Courses

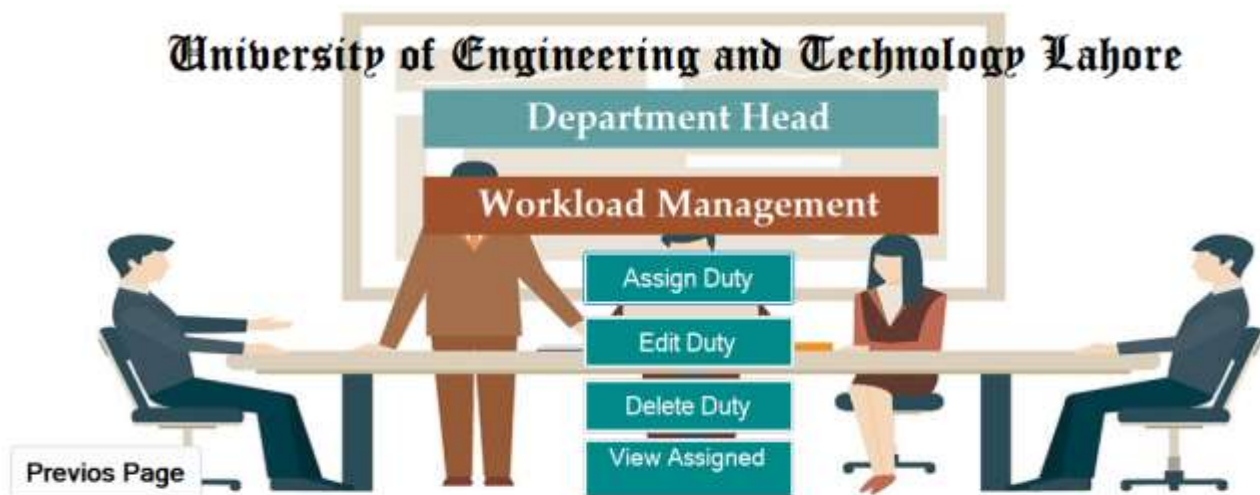
Previous Page

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Assign Course	Button	AssignCourseCrud	Faculty_courses
Edit	Button	AssignCourseCrud	Faculty_courses
Delete Course Assignment	Button	AssignCourseCrud	Faculty_courses
View Assigned Course	Button	ViewCourses	Faculty_courses
Previous Page	Button		

Interface Name

Administrative Duties Assignment

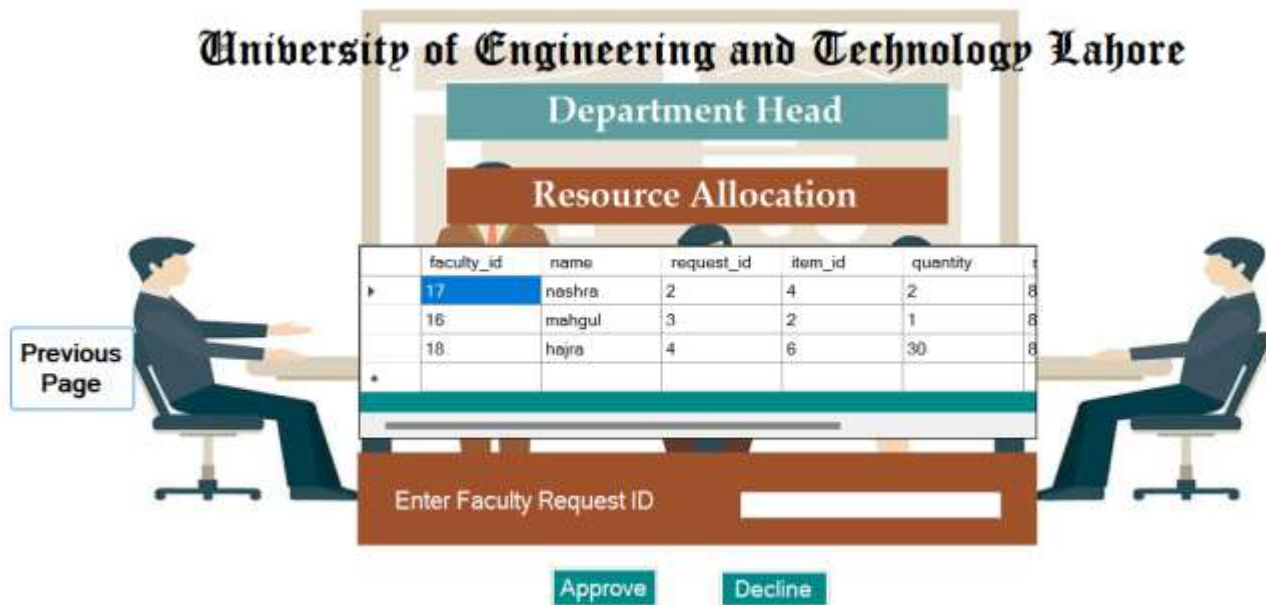
Picture



UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Assign Duty	Button	AdminRoleCrud	Faculty_admin_roles
Edit Duty	Button	AdminRoleCrud	Faculty_admin_roles
Delete Duty	Button	AdminRoleCrud	Faculty_admin_roles
View Assigned Duty	Button	ShowAllRoles	Faculty_admin_roles
Previous Page	Button		

Interface Name	Request Approval/Decline
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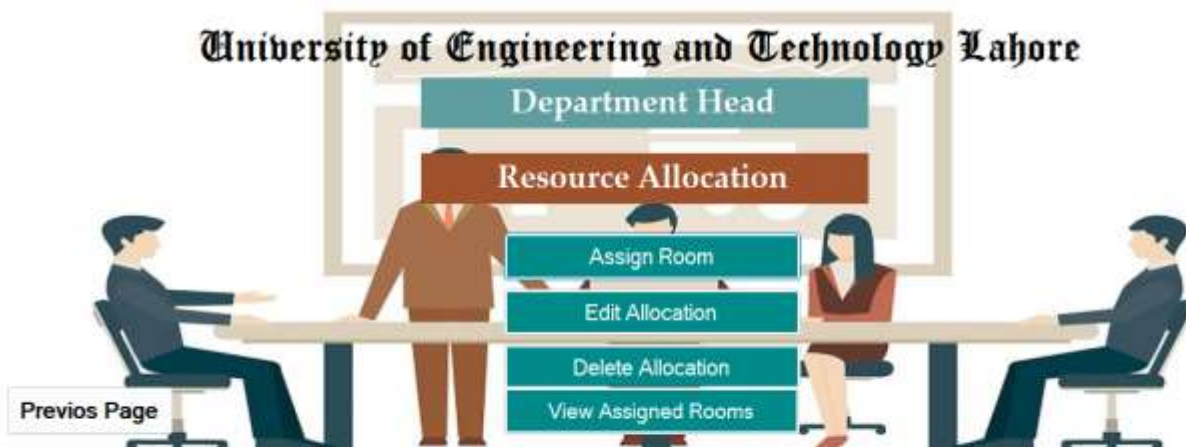
Picture



UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Requests View	DatagridView	ShowReq	Faculty_requests
Approve	Button	Button1_click	Faculty_requests
Decline	Button	Button2_click	Faculty_requests

Interface Name	Room Allocation
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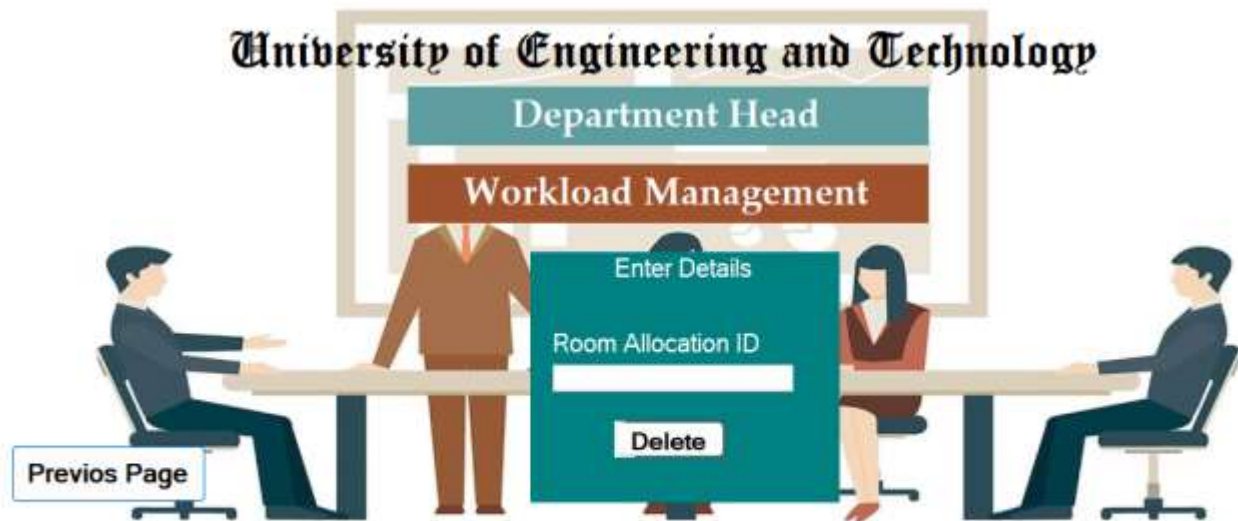
Picture



UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Assign Room	Button	RoomCrud	faculty_room_allocations
Edit Allocation	Button	RoomCrud	faculty_room_allocations
Delete Allocation	Button	RoomCrud	faculty_room_allocations
View Assigned Rooms	Button	RoomCrud	faculty_room_allocations

Interface Name	Delete Room Allocation
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Picture



UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Room Allocation ID	Textbox	RoomCrud	Faculty_room_allocations
Delete	Button	RoomCrud	Faculty_room_allocations
Previous Page	Button		

Interface Name	Report and Analytics
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Picture

University of Engineering and Technology Lahore

Faculty Member

Request Submission

Enter Details

Request ID

Previous Page

Delete

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Request Deletion	button	ReqCrud	Faculty_requests
Request Id	Textbox	ReqCrud	Faculty_request
Previous Page	button		

Interface Name

Request Submission

Picture

University of Engineering and Technology Lahore

Faculty Member

Request Submission

Add Request

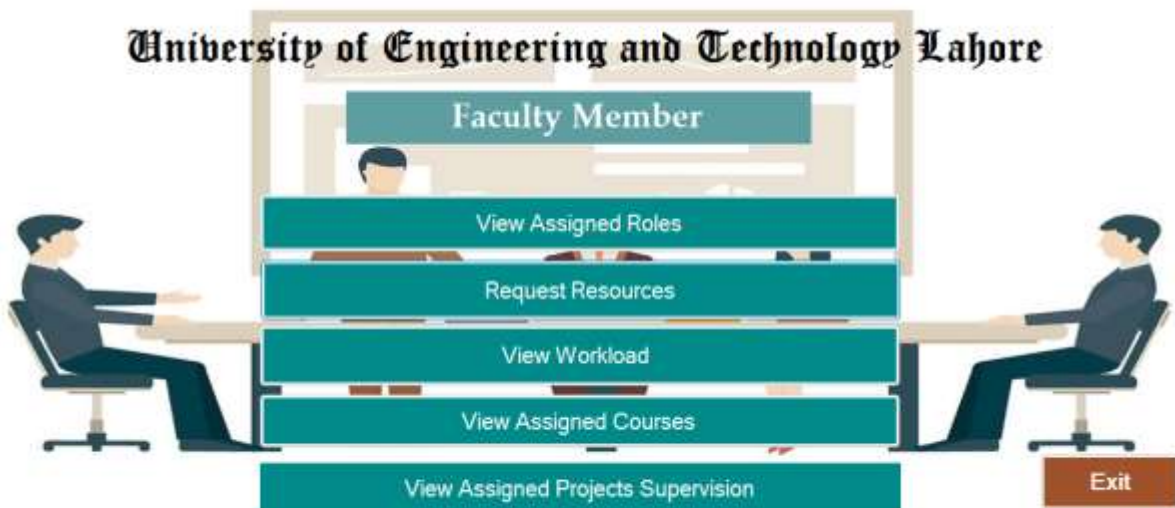
Edit a Request

Delete a Request

View Request Status

Previous Page

UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Add Request	button	ReqCrud	Faculty_requests
Edit Request	button	ReqCrud	Faculty_requests
Delete Request	button	ReqCrud	Faculty_requests
View Request Status	button	ShowStatus	Faculty_requests

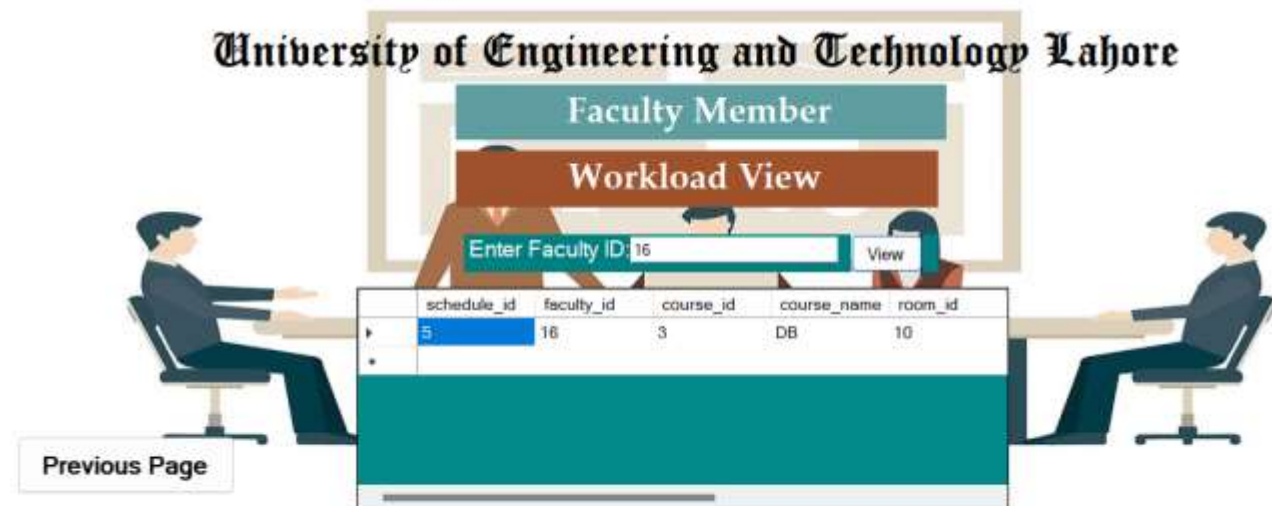
Interface Name		Faculty Member UI	
Picture			
			
UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
View Assigned Roles	Button	FacReqStat	SELECT faculty_admin_roles.admin_role_id, faculty_admin_roles.faculty_id, faculty.name, faculty_admin_roles.role_name, faculty_admin_roles.semester_id FROM Faculty_admin_roles JOIN faculty ON faculty_admin_roles.faculty_id = faculty.faculty_id WHERE faculty_admin_roles.faculty_id= {faculty_id}";
Request Resources	Button	FacreqSub	
View Workload	Button	Workload view	SELECT faculty_course_schedule.schedule_id, faculty_courses.faculty_id, faculty_courses.course_id,courses.course_name, faculty_course_schedule.roo

			m_id, rooms.room_name, faculty_course_schedule.day_ of_week, faculty_course_schedule.start _time, faculty_course_schedule.end_ time FROM faculty_course_schedule JOIN faculty_courses ON faculty_course_schedule.facul ty_course_id = faculty_courses.faculty_cours e_id JOIN courses ON faculty_courses.course_id = courses.course_id JOIN rooms ON faculty_course_schedule.roo m_id = rooms.room_id WHERE faculty_courses.faculty_id = {faculty_id}";
View Assigned Courses	Button	FacProfile	"SELECT faculty_courses.faculty_id, faculty_courses.course_id,cou rses.course_name, courses.course_type, courses.credit_hours,courses.c ontact_hours FROM faculty_courses JOIN courses ON faculty_courses.course_id = courses.course_id WHERE faculty_courses.faculty_id = {faculty_id}";
View Assigned Project Supervision	Button	ProjectSup	"SELECT faculty.faculty_id,faculty.nam e AS faculty_name, faculty_projects.project_id, projects.title AS project_title, projects.description AS project_description, faculty_projects.semester_id,f

			aculty_projects.supervision_h ours FROM faculty JOIN faculty_projects ON faculty.faculty_id = faculty_projects.faculty_id JOIN projects ON faculty_projects.project_id = projects.project_id WHERE faculty.faculty_id ={faculty_id}";
Exirt	Button		

Interface Name	Workload View
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Picture

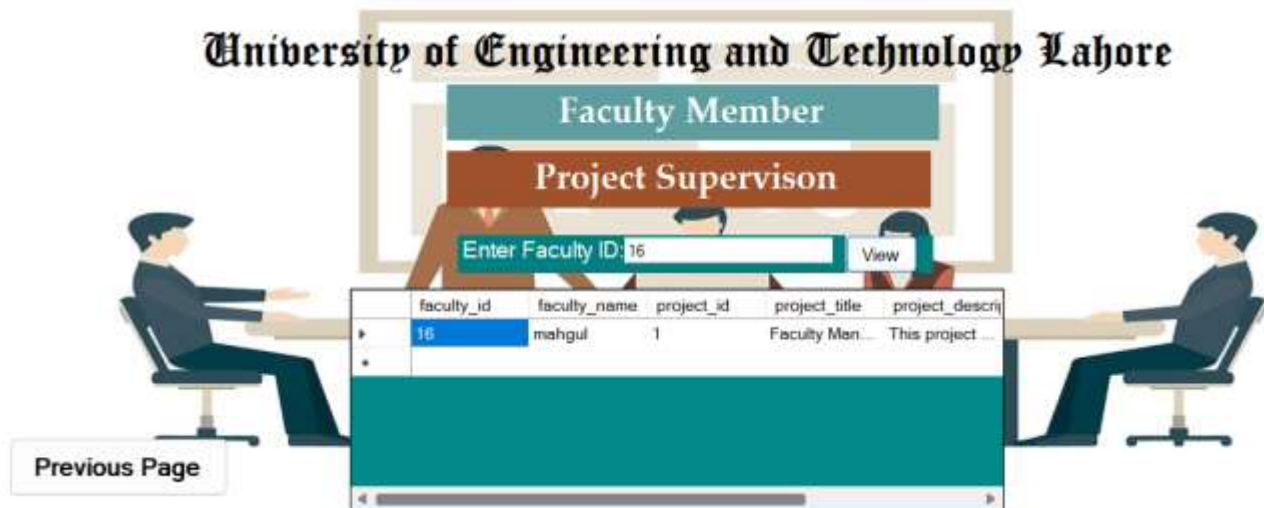


UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Workload view	button	WorkloadView	SELECT faculty_course_schedule.sche dule_id, faculty_courses.faculty_id, faculty_courses.course_id,cou rses.course_name, faculty_course_schedule.roo m_id, rooms.room_name,

			faculty_course_schedule.day_of_week, faculty_course_schedule.start_time, faculty_course_schedule.end_time FROM faculty_course_schedule JOIN faculty_courses ON faculty_course_schedule.faculty_course_id = faculty_courses.faculty_course_id JOIN courses ON faculty_courses.course_id = courses.course_id JOIN rooms ON faculty_course_schedule.room_id = rooms.room_id WHERE faculty_courses.faculty_id = {faculty_id}";
Faculty_id	Textbox		

Interface Name	Project Supervision
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Picture



UI Component Name	Type of UI component	Class Attribute name, Classname, attribute name	Linked to which database column/table
Assigned Project Supervision	DataGridView	ProjectSup	"SELECT faculty.faculty_id,faculty.name AS faculty_name,

			faculty_projects.project_id, projects.title AS project_title, projects.description AS project_description, faculty_projects.semester_id,f aculty_projects.supervision_h ours FROM faculty JOIN faculty_projects ON faculty.faculty_id = faculty_projects.faculty_id JOIN projects ON faculty_projects.project_id = projects.project_id WHERE faculty.faculty_id ={faculty_id}";
Faculty_id	Textbox		

Business Report-1	
<i>Business Report Name: Faculty Workload Details</i>	
<i>Sample of report:</i>	

Faculty Workload Details

schedule id	faculty id	name	course name	room id	room name	day of week	start time	end time
1	9	ayan	PF	10	N10	Monday	08:00:00	11:00:00
2	12	hareem	PFL	1	N1	Tuesday	09:00:00	11:00:00
3	18	hajra	OOPL	6	N6	Wednesday	10:00:00	11:00:00
4	9	ayan	PF	2	N2	Wednesday	01:00:00	04:00:00
5	16	mahgul	DB	10	N10	Friday	01:00:00	04:00:00

Query to build Report: "SELECT faculty_course_schedule.schedule_id, faculty.faculty_id,faculty.name AS faculty_name,courses.course_name,faculty_course_schedule.room_id, rooms.room_name, faculty_course_schedule.day_of_week, faculty_course_schedule.start_time, faculty_course_schedule.end_time FROM faculty_course_schedule JOIN faculty_courses ON faculty_course_schedule.faculty_course_id = faculty_courses.faculty_course_id JOIN faculty ON faculty_courses.faculty_id = faculty.faculty_id JOIN courses ON faculty_courses.course_id = courses.course_id JOIN rooms ON faculty_course_schedule.room_id = rooms.room_id";

Business Report-2

Business Report Name: Faculty Users Details

Sample of report:

Faculty Members Info

User id	Username	Email	Password	Contact	Designation id	Research area	Teaching	Faculty id	Roll id
2	ayan	ayan13@gmail.com	8ecb5bcd8cd84cc3f6cf5dc3076d81c0a457a6bd4b305a33f318b623d701c2e	0123-13333333	5	none	5	9	2
7	hareem	hareem18@gmail.com	9589262630775d921be15b9b2d36fa40f91af0beeb087dae0c721f3c787b2	0123-8888888	6	None	6	12	2
4	mahgul	mahgul23@gmail.com	61503890505f84b144e6ac89124540a3eb8d22e77db76500984cfc50a1d8776e	0123-232323	7	Web Design	4	16	2
6	nashra	nashra27@gmail.com	6195d146a1ee8bd984140dc3c3e6d772765010745db6c311dd0736ddbdc564cca	0123-272727	6	Web Design	5	17	2
8	haga	haga16@gmail.com	ee09186a46224875c139928511af2b855895c43ee86d1da894ae82bc7c990afa	0123-161616	7	Artificial Intelligence	4	18	2
11	areeba	areeba18@gmail.com	8ffe6459134b46975ecd31df13a50c51dbeac1fc19a764bf1602ba7c73fc8f	0123-181818	6	Artificial Intelligence	5	19	2

Query to build Report: *SELECT users.username,users.password_hash AS password, users.email, users.role_id, faculty.contact, faculty.research_area, faculty.total_teaching_hours , faculty.designation_id, users.user_id,faculty.faculty_id FROM faculty JOIN users ON faculty.user_id = users.user_id*

Business Report-3

Business Report Name: Faculty Requests Details

Sample of report:

Faculty Requests Details

request id	faculty id	item id	quantity	status id	request date
2	17	4	2	8	10/03/2025 9:21:40 am
3	16	2	1	8	04/02/2025 7:41:40 am
4	18	6	30	8	19/03/2025 8:47:06 pm
6	9	1	10	10	19/03/2025 4:03:20 am
7	19	7	15	11	12/03/2025 9:40:07 am

Query to build Report: *SELECT faculty_requests.request_id,faculty.faculty_id,faculty_requests.item_id, faculty_requests.quantity, faculty_requests.status_id, faculty_requests.request_date FROM faculty_requests JOIN faculty ON faculty_requests.faculty_id = faculty.faculty_id*

Business Report-4

Business Report Name: Courses Assigned to Faculty Members

Sample of report:

Assigned Courses Detail

faculty course id	faculty id	faculty name	course id	course name	semester id
1	9	ayan	1	PF	1
2	12	hareem	2	PFL	1
3	16	mahgul	3	DB	1
4	16	mahgul	4	DBL	1
5	17	nashra	5	OOP	1
6	18	hajra	6	OOPL	1

Query to build Report: *SELECT faculty_courses.faculty_course_id,faculty.faculty_id,faculty.name , faculty_courses.course_id, courses.course_name , faculty_courses.semester_id FROM faculty_courses JOIN faculty ON faculty.faculty_id = faculty_courses.faculty_id JOIN courses ON faculty_courses.course_id = courses.course_id*